

forces - practice problems

name _____

1. A 6.0 kg box has three horizontal forces on it: 18 N right, 10 N left, and 4.0 N right. What is the box's horizontal acceleration?

2. A kitchen scale shows that a textbook has a mass of 1850 g. The book rests on a flat table. What are the force of gravity and the normal force on the book?

3. A textbook rests on a table. The normal force on the book is equal and opposite to the book's weight. Are these two forces a Newton's third law pair?

4. An 80 kg person rides an elevator up to the 12th floor. At the start of the ride, the elevator accelerates upward at 1.5 m/s^2 . What normal force does the floor apply to the person?

5. An elevator is slowing down as it approaches the top floor. At one moment it is moving upward at 3.0 m/s. What is the normal force on a 20 kg box sitting on the elevator floor?

6. At the airport, a bag scale says your backpack weighs 22 pounds. What is the backpack's mass in kilograms, and what is its weight in newtons? ($1 \text{ kg} = 2.2 \text{ lbs}$ on Earth)
7. A 3.0 kg rock is taken to the Moon and to Mars. What is its weight on each world? Use 1.6 m/s^2 for the Moon and 3.711 m/s^2 for Mars.
8. A supply crate is dropped from a helicopter with a small parachute. The crate's weight is 686 N, and the parachute pulls up on it with 196 N. What are the crate's mass and acceleration?
9. A 10 kg box slides down a smooth 30° loading ramp. Assume no friction. Find the force of gravity, the normal force, the component of gravity down the ramp, and the box's acceleration.
10. The page's example found that a 20 kg box on a frictionless 20° hill accelerates at 3.35 m/s^2 . A 60 kg adult on a sled slides down the same frictionless hill. What are the adult's normal force and acceleration?

11. A 1.2 kg camera drone hovers in a gusty wind. Its rotors push it straight up with 15 N, and the wind pushes it sideways with 3.0 N. Find the drone's horizontal and vertical acceleration.
12. A 15 kg crate is lifted by a rope, and it accelerates upward at 1.2 m/s^2 . What is the tension in the rope?
13. A dog pulls forward with 80 N. The leash angles back and up from the dog at 30° above the horizontal. The dog isn't accelerating. What is the total tension in the leash?
14. A 30 kg cart pulls a 10 kg cart with a rope on a frictionless floor. A student pulls the 30 kg cart forward with 80 N. What is the acceleration of the carts, and what is the tension in the rope between them?
15. A 12 kg block on a frictionless table is connected by a rope over a pulley to a hanging 4.0 kg block. What are the acceleration and the rope tension?
16. Two hanging masses are connected by a rope over a frictionless pulley. One mass is 8.0 kg and the other is 5.0 kg. What are the acceleration and the rope tension?

17. An 18 kg box is held at rest on a frictionless 25° ramp by a rope that pulls up along the ramp. What are the tension and the normal force?
18. A 70 kg person stands on a bathroom scale in an elevator. Scales measure the normal force and then divide by 9.8 to display kilograms. As the elevator starts moving, the scale reads 75 kg. What is the elevator's acceleration?
19. A 20 kg sled is pulled across frictionless ice by a rope with 100 N of tension at 30° above the horizontal. What are the normal force and the horizontal acceleration?
20. A 1200 kg car moving at 72 km/h must stop in 50 m. What constant net force is needed to stop it?
21. A 10 kg block on a frictionless 30° ramp is connected by a rope over a pulley to a hanging 6.0 kg block. The system starts from rest, and the hanging block moves down. Find the acceleration and the tension. How long does it take the hanging block to fall 2.0 m?